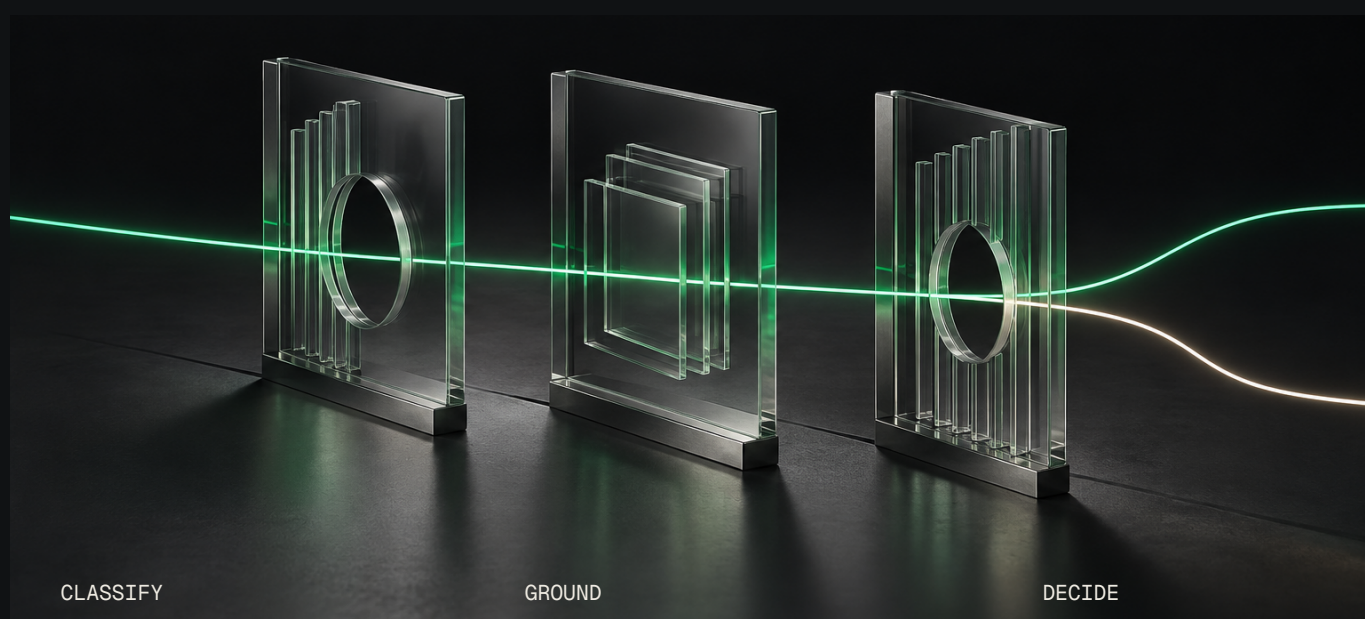


HIVER SDE INTERN TAKE-HOME / 18 SEPTEMBER 2026

An AI support agent that earns its answer.

Built by Arnav Bule. Cadence classifies customer messages, drafts grounded replies and explains when a human must take over.



27,627

Real support threads

SpotifyCares conversations from the 2017 Twitter support corpus.

200

Frozen benchmark cases

Four matched systems. Labels and scores reviewed by Arnav Bule.

80

Completed confirmation cases

Balanced study; AI labels frozen before inference.

The engineering question: How much support can the system usefully handle automatically while missing as few necessary escalations as possible? Fluency and intent accuracy alone do not answer that question.

[Live demo](#)[Repository and runnable evidence](#)

Start with the customer.

A small, inspectable pipeline; explicit labels; separate evidence for each version.

PROBLEM FRAMING

SpotifyCares has recurring technical questions and account-specific requests. “Good” means relevant, supported, useful and appropriately escalated. The agent drafts public replies; it has no account tools and never posts tweets, refunds money or contacts staff.

Twelve intents cover playback, playlists, plans, billing, login, security, downloads, catalog, artist metadata, feedback, language and other requests. Billing/security defaults and deterministic safety rules override model confidence. The threshold is frozen at 0.90.

VERSIONED IMPLEMENTATIONS, EXPLICIT HANDOFFS

DEPLOYED REFERENCE

Rules + BM25 k6 + one structured model call + release checks.

BALANCED CANDIDATE

Joint route + approved answer selection; preserve request scope; check the exact reply in a second call.

HUMAN HANDOFF

Return a reason and a real support contact route. Never claim an action was performed.

ANNOTATION CHAIN

Evidence	Origin and review	Use
Historical 250	AI labels; repeatedly inspected	Training / regression
Frozen 200	AI labels; Arnav approved	Frozen benchmark
100 reply ratings	Astra xhigh; Arnav verified	Judge calibration audit
New 60 + 80	Astra xhigh before inference	Versioned confirmation

Sampling excludes prior candidate, golden, taxonomy and benchmark cases; confirmation also excludes development messages. Conversation overlap and $\geq 85\%$ fuzzy text matches are rejected. This is an English-focused customer-message sample, not traffic-weighted demand.

The headline, with context.

The archived 200-case benchmark is kept immutable. New measurements do not overwrite it.

System	Accuracy	Macro-F1	Recall	Auto	Misses
Majority / escalate	11.5%	0.017	100%	0/200	0
Keyword / nearest	60.0%	0.601	34.6%	166/200	53
Same prompt, kO	80.0%	0.826	100%	0/200	0
Retrieval, k6	77.0%	0.789	93.8%	70/200	5

Read recall together with coverage. Always escalating misses nothing and resolves nothing automatically. The kO arm also has zero coverage because the citation gate vetoes uncited replies; it is not a clean retrieval-safety ablation.

0.789

Intent macro-F1

Paired bootstrap 95% interval: 0.647 to 0.836.

93.8%

Escalation recall

76 / 81 required escalations caught.
Bootstrap interval: 88.2% to 98.8%.

5 / 70

Misses among auto

A fluent automatic response can still require a human.

A HIGH JUDGE SCORE WAS NOT ENOUGH

The original Gemini judge scored agent replies 4.11/5 versus 2.69/5 for the simple baseline, but approved an obsolete limit and a fictional DM action. Preference order consistency was only 73.5%. Retrieval did not establish a classification improvement: k6 - kO accuracy was -3.0 points (95% interval -7.5 to +1.0).

Calibration audit: 100 exact replies on 50 matched messages were first rated by GPT-6 Astra at extra-high reasoning, then verified unchanged by Arnav Bule. Agreement against both Gemini orders: weighted kappa 0.137, exact 28%, within one 53%. The human saw AI scores; the stricter later rubric is a disclosed audit, not an independent blind replication.

Fix the reply, not the score.

Five recurring defects explain the candidate design. Known cases remain regression evidence.

01 Unsupported actions

h_168 claims a DM was sent; d2_003 claims feedback forwarding. Hypothesis: copied staff actions become promises.

Server-rendered actions prevent arbitrary operational promises. Rules and exact-text semantic review add checks.

02 Wrong-issue guidance

d2_007 confuses customer and Spotify-created playlists. Hypothesis: lexical overlap outranks issue fit.

Classify before retrieval. Review whether the exact proposed step addresses the customer's stated issue.

03 Stale procedures

h_003 repeats an old limit; d2_009/010 copy developer-work claims. Hypothesis: history becomes current policy.

Treat history as examples, not current policy. Dated official sources authorize a narrow set of present procedures.

04 Unhelpful handoffs

d2_010/d2_013 give no path forward. Hypothesis: abstention suppresses useful guidance.

Use an honest limitation and Spotify's verified support contact page; ask only necessary, specific questions.

05 Risk and ambiguity

h_079/h_105 expose frustration and churn misses. Hypothesis: coarse confidence masks escalation risk.

Keep rules, primary/secondary policy defaults and low-confidence escalation. A semantic checker cannot prove safety.

Review before release: the semantic check sees the exact finalized public text, including trimming and links. Failed checks produce a contact handoff. BalancedAgent uses one or two calls; tokens and latency are measured rather than assumed.

Measure useful coverage.

Development selects a hypothesis. Fresh confirmation tests the frozen implementation.

BEFORE / THE EARLIER 60-CASE CONFIRMATION

The earlier quality candidate reduced missed escalations from 4 to 0, but automatic replies fell from 22/60 to 4/60. Useful replies rose only from 3/60 to 4/60. That coverage loss motivated a broader set of verified answers and joint routing. The reference and quality versions remain unchanged.

BALANCED / FROZEN 80-CASE STUDY / THREE ARMS / AI REVIEW

System	Auto	Useful auto	Resolution-style	Misses
Reference	28/80	6/80	0	4
Quality candidate	7/80	6/80	1	0
Balanced candidate	30/80	13/80	11	4

COST AND LATENCY / SUCCESSFUL END-TO-END PREDICTIONS

Runtime	Reference	Quality	Balanced
p95 seconds	4.06	4.53	3.86
Input + output tokens	205,542	199,047	386,582

Useful automatic requires ship, grounding/safety/next-step scores ≥ 4 and no flags. Clarifications count; resolution-style replies are reviewer judgments, not observed outcomes. Handoffs/social replies are excluded. Misses count even when text looks harmless. All recorded calls were fresh. Timing spans 2026-09-17, 2026-09-18 (UTC). Failed attempts and interruption time are excluded.

Acceptance: Technical gates failed. Keep the deployed reference; not all promotion gates are met. Failed: missed-escalation bound; zero flagged automatic replies. Balanced automatic replies flagged by review: 4. New labels and reply ratings are AI-authored. Arnav Bule verified the earlier 100 ratings; that verification does not cover these new outputs. Development iterations are preserved. Balanced was frozen before confirmation and remains unchanged.

Make the claim reproducible.

The evidence is worth more than the demo. Reproduce saved results without an API key.

WHAT IS MISLEADING ABOUT MY HEADLINE NUMBER?

Coverage hides utility. Automatic replies can be irrelevant or unsafe; useful coverage and misses matter.

Confidence is not calibrated risk. The original 0.90 threshold used an F2 fallback, not the claimed recall target.

Samples limit the claim. Old AI labels had human review; 60/80-case labels were AI-only. Sampling is not traffic-weighted.

Review is not independent. Arnav verified visible Astra scores; the later stricter rubric changes agreement.

Replay is not a fresh run. Inspected cases are regressions. Successful-call timings are not a production SLA.

VERIFIEDAGENT / COMPLETED PRE-PATCH STUDIES / AI ONLY

VerifiedAgent separates risk from answer selection, checks action prerequisites and uses expiring official sources. Exact replies are audited before release.

Study / policy v2	Auto	Useful*	Miss / need	Flagged auto
Development / 80	31	22	2/28	4
Fresh AI pilot / 100	33	22	2/31	6

Development p95: 5.69s; 347,771 tokens. Pilot p95: 5.28s; 435,306 tokens. *Useful requires an eligible automatic reply, ship and all quality gates. Development reuses inspected cases. Both studies use AI labels and review under a new policy/rubric; they are not directly comparable to page 5.

The pilot exposed harm-routing failures. The follow-up guard's 18 selected cases had 0/7 missed escalations and one useful reply. This is regression evidence, not restored coverage or independent confirmation; the table describes pre-patch code.

WITH ONE MORE WEEK / BEFORE PROMOTION

Complete human labels on 200 representative cases and 80 challenge cases before confirmation inference, then matched blind reply review. Run a canary only if gates pass. Earlier human approvals retain their original scope. The reference remains deployed.

REPRODUCE / PYTHON 3.12 / ZERO NEW MODEL CALLS

```
python -m pip install -e ".[dev]"
python analysis_tools/reproduce_benchmark.py
python analysis_tools/reproduce_balanced.py
python analysis_tools/reproduce_verified.py --experiment results/verified_dev_v4
```

[VerifiedAgent study](#)

[Dataset: Twitter customer support](#)

[Reproduction & decision log](#)

[Spotify: official support contact](#)